



# Mortgage Pricer — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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# 1 Test Results — Mortgage Pricer (MKM-MP-001)

Tests run on **22 March 2026 at 20:17:28**.

Table 1: Test Results Summary — Mortgage Pricer (MKM-MP-001)

|             |       |
|-------------|-------|
| Total Tests | 67    |
| Passed      | 67    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.05s |

Table 2: Detailed Test Results — Mortgage Pricer

| Test Description                                                | Acceptance Criteria                                                                                               | Result | Time   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Default ids assigned                                            | <code>results[0]["mortgage_id"] == "MORTGAGE_0";</code><br><code>results[0]["property_id"] == "PROPERTY_0"</code> | Pass   | 0.001s |
| Error isolation                                                 | <code>len(results) == 3;</code> "error" not in <code>results[0]</code> (+2 more)                                  | Pass   | 0.001s |
| Prices multiple mortgages                                       | <code>len(results) == 2;</code><br><code>results[0]["mortgage_id"] == "M1"</code> (+1 more)                       | Pass   | 0.001s |
| Custom tax rate                                                 | <code>MortgagePricer(tax_rate=0.40).tax_rate == 0.40</code>                                                       | Pass   | 0.000s |
| Default tax rate                                                | <code>MortgagePricer().tax_rate == 0.20</code>                                                                    | Pass   | 0.000s |
| Custom tax rate                                                 | <code>spread_high_tax &gt; spread_low_tax</code>                                                                  | Pass   | 0.000s |
| Floor at 10bps                                                  | <code>spread &gt;= 0.001</code>                                                                                   | Pass   | 0.000s |
| Higher affordability higher spread                              | <code>spread_low_income &gt; spread_high_income</code>                                                            | Pass   | 0.000s |
| Negative income fallback                                        | <code>spread == 0.15</code>                                                                                       | Pass   | 0.000s |
| Spread bounded by schedule                                      | <code>spread &gt;= 0.001; spread &lt; 0.05</code>                                                                 | Pass   | 0.000s |
| Zero income fallback                                            | <code>spread == 0.15</code>                                                                                       | Pass   | 0.000s |
| debug=True in the credit spread calculation.                    | <code>"credit_spread" in result</code>                                                                            | Pass   | 0.001s |
| debug=True exercises logging branches without error.            | <code>result["mortgage_value"] &gt; 0</code>                                                                      | Pass   | 0.001s |
| Zero income with debug=True exercises zero-income debug branch. | <code>result["credit_spread"] &gt; 0</code>                                                                       | Pass   | 0.001s |
| flood_risk_category=None with debug=True.                       | <code>result["mortgage_value"] &gt; 0</code>                                                                      | Pass   | 0.001s |

| Test Description                                                             | Acceptance Criteria                                                                                                                | Result | Time   |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| When interest_rate=0, monthly_rate==0 → loan_amount/n_periods formula.       | result["mortgage_value"] > 0                                                                                                       | Pass   | 0.001s |
| Batch passes flood risk                                                      | results[0]["flood_risk_factor"] == 1.40; results[1]["flood_risk_factor"] == 1.00 (+1 more)                                         | Pass   | 0.001s |
| Case insensitive                                                             | MortgagePricer.calculate_flood_risk_impact("high") =...;<br>MortgagePricer.calculate_flood_risk_impact("VERY LOW") ==... (+1 more) | Pass   | 0.000s |
| Flood factor in result                                                       | result["flood_risk_factor"] == 1.20                                                                                                | Pass   | 0.001s |
| Flood risk increases spread                                                  | high["credit_spread"] > low["credit_spread"];<br>high["flood_risk_factor"] > low["flood_risk_factor"] (+1 more)                    | Pass   | 0.001s |
| Flood risk multipliers[high-1.4] [category='High', expected=1.4]             | MortgagePricer.calculate_flood_risk_impact("High") == e...                                                                         | Pass   | 0.000s |
| Flood risk multipliers[low-1.05] [category='Low', expected=1.05]             | MortgagePricer.calculate_flood_risk_impact("Low") == e...                                                                          | Pass   | 0.000s |
| Flood risk multipliers[medium-1.2] [category='Medium', expected=1.2]         | MortgagePricer.calculate_flood_risk_impact("Medium") == e...                                                                       | Pass   | 0.000s |
| Flood risk multipliers[very high-1.75] [category='Very High', expected=1.75] | MortgagePricer.calculate_flood_risk_impact("Very High") == e...                                                                    | Pass   | 0.000s |
| Flood risk multipliers[very low-1.0] [category='Very Low', expected=1.0]     | MortgagePricer.calculate_flood_risk_impact("Very Low") == e...                                                                     | Pass   | 0.000s |
| No flood risk defaults to unity                                              | result["flood_risk_factor"] == 1.0                                                                                                 | Pass   | 0.001s |
| None returns unity                                                           | MortgagePricer.calculate_flood_risk_impact("None") == 1.0                                                                          | Pass   | 0.000s |
| Unknown category returns unity                                               | MortgagePricer.calculate_flood_risk_impact("Unknown") == 1.0;<br>MortgagePricer.calculate_flood_risk_impact("Extreme") == 1.0      | Pass   | 0.000s |
| Ltv thresholds[0.5-1.0] [ltv=0.5, expected=1.0]                              | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |
| Ltv thresholds[0.8-1.0] [ltv=0.8, expected=1.0]                              | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |
| Ltv thresholds[0.85-1.1] [ltv=0.85, expected=1.1]                            | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |

| Test Description                                  | Acceptance Criteria                                                                                 | Result | Time   |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Ltv thresholds[0.9-1.1] [ltv=0.9, expected=1.1]   | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.91-1.3] [ltv=0.91, expected=1.3] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.95-1.3] [ltv=0.95, expected=1.3] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.96-1.5] [ltv=0.96, expected=1.5] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[1.0-1.5] [ltv=1.0, expected=1.5]   | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Zero property value                               | <code>pricer.calculate_loan_to_value_impact(100.0, 0) == 1.5</code>                                 | Pass   | 0.000s |
| Higher rate higher payment                        | <code>high &gt; low</code>                                                                          | Pass   | 0.000s |
| Shorter term higher payment                       | <code>short &gt; long</code>                                                                        | Pass   | 0.000s |
| £200k at 3.5% over 25yr<br>£1,001.25/month.       | <code>995 &lt; payment &lt; 1010</code>                                                             | Pass   | 0.000s |
| Zero-rate degenerate case: $M = P / n$ .          | <code>payment == pytest.approx(1000.0)</code>                                                       | Pass   | 0.000s |
| Each mortgage has property id                     | <code>mapping.get("property_id") is not None</code>                                                 | Pass   | 0.003s |
| Processing stats                                  | <code>stats["successful_mortgages"] == 5;</code><br><code>stats["failed_mortgages"] == 0</code>     | Pass   | 0.003s |
| Output file exists                                | <code>result["file_path"].exists()</code>                                                           | Pass   | 0.003s |
| Output file is valid json                         | <code>isinstance(data, dict)</code>                                                                 | Pass   | 0.005s |
| Generate returns expected keys                    | <code>"data" in result;</code> <code>"file_path" in result (+1 more)</code>                         | Pass   | 0.003s |
| Mortgage count matches properties                 | <code>len(result["data"]["mortgages"]) == 5</code>                                                  | Pass   | 0.003s |
| Mortgage id format                                | <code>re.match(r"~MORT-[a-f0-9]{8}\$", mid)</code>                                                  | Pass   | 0.003s |
| Mortgage ids unique                               | <code>len(ids) == len(set(ids))</code>                                                              | Pass   | 0.003s |
| All errors returns error                          | <code>"error" in calculate_portfolio_metrics(results)</code>                                        | Pass   | 0.000s |
| High risk count                                   | <code>calculate_portfolio_metrics(results)["high_risk_mortgages"]</code>                            | Pass   | 0.000s |
| Mixed results exclude errors                      | <code>metrics["total_mortgages"] == 1;</code><br><code>metrics["error_count"] == 1</code>           | Pass   | 0.001s |
| Valid results                                     | <code>metrics["total_mortgages"] == 2;</code><br><code>metrics["error_count"] == 0 (+3 more)</code> | Pass   | 0.001s |
| Balance converges to zero                         | <code>result["outstanding_balance"][-1] &lt; 1.0</code>                                             | Pass   | 0.001s |
| Expected losses non negative                      | <code>all(e1 &gt;= 0 for e1 in result["expected_losses"])</code>                                    | Pass   | 0.001s |

| Test Description                    | Acceptance Criteria                                                                                                      | Result | Time   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Fair value less than par            | <code>result["mortgage_value"] &lt; base.mortgage_params["loan_amo...; result["discount_to_par"] &gt; 0 (+1 more)</code> | Pass   | 0.001s |
| Hazard rates positive               | <code>result["hazard_rates"][0] == 0.0; all(result["hazard_rates"][i] &gt; 0 for i in range(1, len(r...</code>           | Pass   | 0.001s |
| High ltv increases spread           | <code>high_ltv["credit_spread"] &gt; low_ltv["credit_spread"]; high_ltv["ltv_factor"] &gt; low_ltv["ltv_factor"]</code>  | Pass   | 0.001s |
| Higher haircut more loss            | <code>high["pv_losses"] &gt; low["pv_losses"]</code>                                                                     | Pass   | 0.001s |
| Input validation clamps             | <code>result["mortgage_value"] &gt; 0</code>                                                                             | Pass   | 0.000s |
| Ltv factor in result                | <code>"ltv_factor" in result; result["ltv_factor"] == 1.0</code>                                                         | Pass   | 0.001s |
| Raises on invalid types             | —                                                                                                                        | Pass   | 0.000s |
| Recovery haircut zero low ltv       | <code>all(lgd == 0 for lgd in result["lgds"])</code>                                                                     | Pass   | 0.001s |
| Survival first period less than one | <code>result["survival_probs"][1] &lt; 1.0</code>                                                                        | Pass   | 0.001s |
| Survival monotonically decreasing   | <code>surv[i] &lt;= surv[i - 1]</code>                                                                                   | Pass   | 0.001s |
| Positive rate exceeds principal     | <code>cost &gt; 200.000</code>                                                                                           | Pass   | 0.000s |
| Zero rate equals principal          | <code>cost == pytest.approx(100_000.0)</code>                                                                            | Pass   | 0.000s |

## 1.1 Test Document History

| Date        | Version | Description                                      | Author   |
|-------------|---------|--------------------------------------------------|----------|
| 05-Mar-2026 | 1.0     | Initial test suite (65 tests)                    | D. Kelly |
| 07-Mar-2026 | 1.1     | 62 test(s) added; 65 test(s) removed (total: 62) | D. Kelly |
| 22-Mar-2026 | 1.2     | 5 test(s) added (total: 67)                      | D. Kelly |